

The Effects of Content-Aware Textured Reverse Motion Flow on Cybersickness and User Experience

ByeongSun Hong *
DeepXRLab Inc, Seoul,
Korea, Republic of

Qimeng Zhang †
Korea University, Seoul,
Korea, Republic of

Gerard Jounghyun Kim ‡
Korea University, Seoul,
Korea, Republic of

Jun Ryu§
Korea University, Seoul,
Korea, Republic of

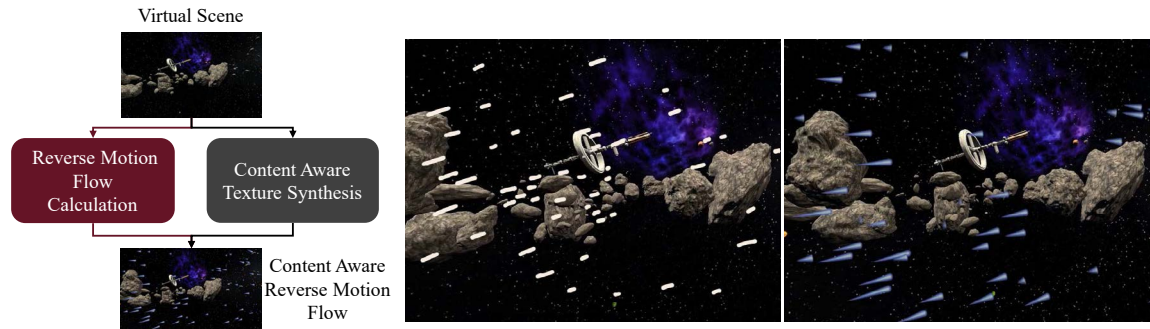


Figure 1: Flowchart of the proposed content aware motion flow augmentation (left), the original reverse optical flow visualization with white moving dots [4] (center), and the proposed visualization with content harmonious texture/color (right)

ABSTRACT

Difficulty arises in virtual reality (VR) due to cybersickness. The discrepancy in motion information between the visual and vestibular feedback causes sensory conflict, a widely accepted cause of cybersickness. Previous research has shown reduced VR sickness by augmenting the content with motion patterns in the opposite direction of the virtual motion. However, it can also cause significant content intrusion. This poster aims to mitigate the latter by applying content-aware textured motion patterns vs. simple white animated feature points. Positive results were obtained with a higher preference for the proposed visualization while showing a similar degree of sickness reduction.

Index Terms: Artificial intelligence, Cybersickness, Image captioning, Optical flow, Simulator sickness, Virtual reality.

1 INTRODUCTION

Various visual techniques have been deployed in virtual reality environments to reduce cybersickness. A crucial aspect to consider when applying cybersickness mitigation methods is that the additional visual elements used should mitigate sickness without compromising the user's sense of immersion and content understanding.

As such cybersickness reduction approaches often adopt visual effects in the peripheral area - e.g. blurring or reduction - in an attempt to reduce the effect of visual motion [5, 2]. A method that mixes in the reverse motion flow (or optical flow) to partially cancel out the visual motion, and thereby reduce motion sickness, has also been proposed [4]. However, the current implementation uses lines or simple animated feature points that do not blend well with the

given scene, disrupting the sense of immersion and user experience [3]. To address this issues, this poster proposes a novel approach that automatically recommends textures and colors (to be applied to the visualization of the reverse optical flow (ROF)) that is in harmony with the given virtual environment (as shown in Figure 1).

2 TEXTURED REVERSE OPTICAL FLOW VISUALIZATION

Off-line Scene Analysis and Texture Synthesis

First, virtual environments are categorized by themes (e.g., technology, nature, city, etc.), and a texture database specific to each category is pre-constructed. We define four broadly applicable databases: Space, Nature, Speed, and Tech, each containing 3–5 transparent, colorless textures. For instance, the theme of “Space” might include texture shapes of meteor trails and nebulae (likewise, “Nature” features would include leaves and wind patterns, and for “Speed”, motion blurs and water droplets), enabling flexible application across diverse scenes. It is essential to ensure that the selected textures, when applied, do not compromise the user's sense of immersion.

Texture/Color Selection and Particle based Visualization

For the virtual environment at hand, 30 scene frames are randomly extracted from its content. GPT-4o-based neural image captioning method [1] generates descriptive sentences for these frames, which are then analyzed using text mining techniques to extract key textual features. Subsequently, the text features are matched with the most appropriate textures from the pre-constructed database. Simultaneously, the GPT-4o-based color extraction identifies the dominant color from the scene frames, which is most suitable for the environment as guided by the color theory (we omit further details). The texture with the selected color is to be applied to the augmented motion particles (see next section) to maintain as much visual harmony with the main content. Figure 2 illustrates the overall process.

In the original version [4], the ROF was generated from the screen space moving features, a significant computational burden on real-time rendering. A 3D particle system emulating the movement in the direction of the user's motion (seen as moving oppo-

*e-mail: bshong@deepxrlab.com

†e-mail: zoe1024@korea.ac.kr

‡e-mail: gjkim@korea.ac.kr

§e-mail: rjstarwars@korea.ac.kr

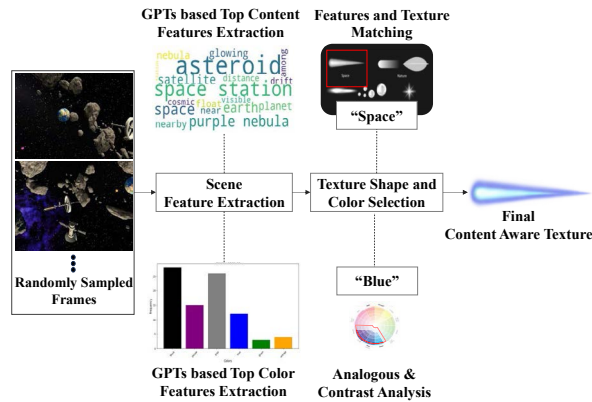


Figure 2: The process of selecting content-aware texture by extracting key descriptive words by image captioning and using them to select the proper texture shape and color from the database.

site to the natural optical flow) was used instead. For example, a player moving forward would see a natural radial form of optical flow heading backward. The particle system in this case would augment a forward radial flow, in the same direction as the user's virtual movement. The selected textures and colors, explained in the previous section, are applied to the particles. Figure 1 (right) depicts the application of the texture/color to the particles.

3 EXPERIMENT

The experiment was run to test the effect of the textured ROF visualization. It was designed as a balanced Latin square design with 1×4 within-subject single measure. The factor being the texture being applied and the direction of the particles: (1) without any visual manipulation (as is, Default), (2) with the selected texture applied and particles moving in the same direction as the original reverse optical flow (ROF-T-F, F stands for forward), (3) with the selected texture applied and particles moving in the opposite direction as the original reverse optical flow (ROF-T-B, B stands for backward), (4) the original (visualized with simple animated feature points) Reverse Optical Flow (ROF-OG).

Participants were asked to input their Fast Motion Sickness score by turning a dial during the experiment. 28 subjects (age of 19 and 41, $M = 24.27$, $SD = 4.33$) were presented with four treatments with only one treatment applied per day to minimize the carry-over effect of cybersickness. Each treatment lasted around 5 minutes. After each treatment the participants were asked to fill out the Fast Motion Sickness Scale questionnaire (FMS), Kennedy's Simulation Sickness Questionnaire, and a presence/immersion survey. After all treatments, a survey for preference and the perceived motion among all treatments was also administered.

4 RESULTS

The proposed method of utilizing content-aware reverse motion flow (ROF-T-F) has shown promising results in reducing cybersickness and preference. FMS score results are as shown in Figure 3 with the Default condition giving ($M = 9.035$, $SD = 3.494$), ROF-T-F ($M = 7.035$, $SD = 4.122$), ROF-T-B ($M = 8.428$, $SD = 4.220$), ROF-OG ($M = 7.571$, $SD = 5.398$). Repeated Measures ANOVA test results did not show statistical significance. ($p = .104$).

Other questionnaire results showed a similar trend with the ROF-T-F showing the best results in reducing cybersickness, however, Repeated Measures ANOVA results were either not statistically significant or failed to meet the normality assumptions (Shapiro-Wilk test). From the post-experiment questionnaires, ROF-T-F was the

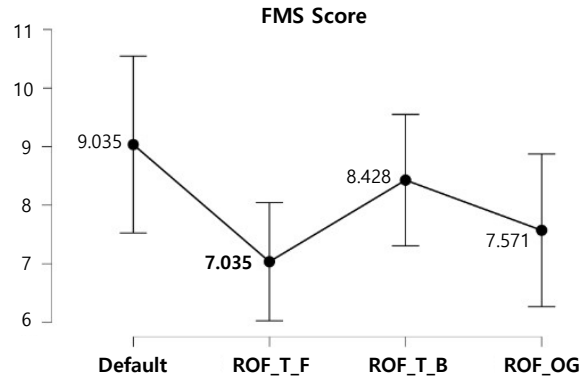


Figure 3: Fast Motion Sickness Questionnaire results show the lowest average self-reported sickness rating on the proposed method (ROF-T-F)

most preferred condition (11 votes among 27 voters) and also the last in the least preferred condition category (5 votes among 27 voters).

5 CONCLUSION

We have developed a novel method of generating and applying content-harmonious textures to the ROF pattern to mitigate cybersickness. The proposed method showed a reduction in cybersickness, comparable to the original approach, but was much preferred by its less content-intrusive nature. A major limitation lies in the fact that only a single texture based on the theme of the virtual space is used. For more dynamic scenes, the texture/color should change accordingly to maintain harmony with the content. Future work aims to improve the heuristic methods of augmenting the reverse motion flow while looking more in-depth at how the direction of the flow and the augmented texture affect user experience and cybersickness. The improved version of this method will be tested with additional subjects to improve the statistical power.

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REFERENCES

- [1] J. Achiam, S. Adler, S. Agarwal, L. Ahmad, I. Akkaya, F. L. Aleman, D. Almeida, J. Altschmidt, S. Altman, S. Anadkat, et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023. 1
- [2] H. Buhler, S. Misztal, and J. Schild. Reducing vr sickness through peripheral visual effects. In *2018 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pp. 517–9, 2018. doi: 10.1109/VR.2018.8446346 1
- [3] S. Kim and G. J. Kim. Dynamically adjusted and peripheral visualization of reverse optical flow for vr sickness reduction. *Electronics*, 12(4), 2023. doi: 10.3390/electronics12040861 1
- [4] S. H. Park, B. Han, and G. J. Kim. Mixing in reverse optical flow to mitigate vection and simulation sickness in virtual reality. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*, CHI '22. Association for Computing Machinery, New York, NY, USA, 2022. doi: 10.1145/3491102.3501847 1
- [5] J.-h. Won and Y. S. Kim. A study on visually induced vr reduction method for virtual reality sickness. *Applied Sciences*, 11(14), 2021. doi: 10.3390/app11146339 1